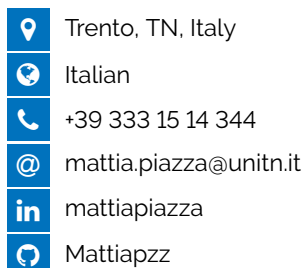


# MATTIA PIAZZA

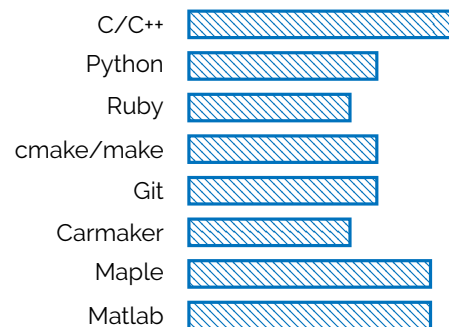
Researcher  
Mechatronics Engineer



## WHO AM I?

I am a motivated and results-driven mechatronics engineer with a passion for autonomous driving, optimization, and control systems. Currently a doctoral researcher at the University of Trento, I focus on optimal control algorithms for ADAS in collaboration with leading automotive and motorsport partners. In my journey, I designed and implemented a prototype control system and a real-time, multiplatform library. Throughout my career, I have gained strong expertise in C++ programming, symbolic computation, and optimal control, with additional experience in cross-platform compilation, low-level software interfaces, and communication protocols (UDP, TCP/IP). Furthermore, I have hands-on experience with multi-sensor software frameworks (e.g., RTMaps) for hardware-in-the-loop and real-vehicle testing—bridging theoretical research with practical application. I bring curiosity, ambition, and determination to every challenge I take on.

## SOFTWARE AND PROGRAMMING SKILLS



## WORKING EXPERIENCES

- 2022–Present **PhD Researcher - Doctoral program in Materials, Mechatronics and Systems Engineering** University of Trento  
Optimal Control, Motion Planning, Connected and Automated Vehicles and ADAS systems.
- 2021–2022 **Research Fellowship (Research grant decree n. 205/2020)** University of Trento  
Title: "Solution algorithms comparison for minimum-time optimal control problems for racing vehicles".
- 2020–2021 **Research scholarship (Decree n. 74/2020)** University of Trento  
Title: "Minimum Time Manoeuvres of complex vehicle models described with DAE".
- 2017 **Internship in Quality Control department** SIAP s.p.a. gruppo CARRARO
- 2012 & 2013 **Plumber apprentice** DE NARDO ALESSANDRO IMPIANTI TERMOIDRAULICI

## PROJECTS WORKING ON

- 2024–Present **SUNRISE** (Safety assurance framework for connected, automated mobility systems) European Project | University of Trento  
Horizon Europe project to develop an harmonized and scalable CCAM Safety Assurance Framework according to automotive stakeholders needs.
- 2022–2024 **VeDi 2025 - Ministero Italiano dello Sviluppo Economico** CRF - STELLANTIS | University of Trento  
Collaboration with CRF within the VeDi 2025 project financed by MISE. Contribution: development of a motion planner able to negotiate maneuvers for autonomous vehicles equipped with ADAS systems.
- 2021–Present **Minimum Lap Time Simulator** APRILIA RACING | University of Trento  
Collaboration with Aprilia Racing team to develop models and algorithms to solve optimal control problem for minimum lap time offline analysis. This work is a spin-off of my master thesis.
- 2022–2023 **Automatic generation of racetracks for vehicle simulators** AnteMotion | University of Trento  
Collaboration with AnteMotion for automatic generation of racetracks and scenarios for vehicle simulators.

## LANGUAGES

|                         | Listening | Reading | Spoken Interaction | Spoken Production | Writing |
|-------------------------|-----------|---------|--------------------|-------------------|---------|
| Italian (mother tongue) | C2        | C2      | C2                 | C2                | C2      |
| English                 | C1        | C1      | C1                 | C1                | C1      |
| German                  | A1        | A1      | A1                 | A1                | A1      |

## EDUCATION

|              |                                                                                                                                                                                    |                                             |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 2025–Present | <b>Visiting Researcher - Autonomous Vehicle Systems Laboratory</b><br>Optimal control and velocity profile optimization for INDY Autonomous Challenge.                             | Technische Universität München (TUM)        |
| 2022–Present | <b>PhD Candidate - Doctoral program in Materials, Mechatronics and Systems Engineering</b><br>Optimal Control, Motion Planning, Connected and Automated Vehicles and ADAS systems. | University of Trento                        |
| 2020         | <b>Professional qualification to practice as an engineer - Industrial Engineer</b>                                                                                                 | University of Trento                        |
| 2017–2020    | <b>Master of Science degree in Mechatronics Engineering - Mechanics e Mechatronics</b>                                                                                             | University of Trento                        |
| 2013–2017    | <b>Bachelor of Science degree in Mechanical Engineering</b>                                                                                                                        | University of Udine                         |
| 2008–2013    | <b>Scientific High school diploma</b>                                                                                                                                              | Liceo Scientifico "Michelangelo Grigoletti" |

## TEACHING EXPERIENCES

|              |                                                                                                                                                                                                                       |                      |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 2023–Present | <b>Teaching assistant – Mechatronics Systems Simulation</b><br>Teaching assistant for courses of <i>Mechatronics Systems Simulation</i> (Prof. F. Biral and Prof. E. Bertolazzi).                                     | University of Trento |
| 2022–Present | <b>Tutoring and mentoring of thesis students</b><br>Tutoring, managing and mentoring of multiple students during their research thesis with Prof. F. Biral.                                                           | University of Trento |
| 2021–Present | <b>Teaching assistant – Sistemi Meccanici e Modelli and Fondamenti di Meccanica</b><br>Teaching assistant for courses of <i>"Sistemi Meccanici e Modelli"</i> and <i>"Fondamenti di Meccanica"</i> (Prof. M. Da Lio). | University of Trento |
| 2018–2019    | <b>Tutorship specific areas for mechanics and mechatronics and video-lectures</b><br>Frontal lecturer and video lecture (DII Youtube channel) for MSc in Mechatronics Engineering.                                    | University of Trento |

## PUBLICATIONS

- [1] M. Piccinini, S. Taddei, M. Larcher, M. Piazza, and F. Biral, "A physics-driven artificial agent for online time-optimal vehicle motion planning and control," *IEEE Access*, 2023.
- [2] M. Piazza, M. Piccinini, S. Taddei, and F. Biral, "Mptree: A sampling-based vehicle motion planner for real-time obstacle avoidance," *IFAC-PapersOnLine*, vol. 58, no. 10, pp. 146–153, 2024.
- [3] M. Piccinini, S. Taddei, M. Piazza, and F. Biral, "Impacts of ggv constraints formulations on online minimum-time vehicle trajectory planning," *IFAC-PapersOnLine*, vol. 58, no. 10, pp. 87–93, 2024.
- [4] M. Piazza, E. Bertolazzi, and M. Frego, "A non-smooth numerical optimization approach to the three-point dubins problem (3pdp)," *Algorithms*, vol. 17, no. 8, p. 350, 2024.

## THESIS

- [1] M. Piazza, "Influence of Slip Versus Torque Control Formulation on Minimum Time Manoeuvres of a Motorcycle", Master of Science thesis, Supervisor: Prof. F. Biral, University of Trento, Italy, 2020.

## HOBBIES AND SPORTS

Outside of work, I am passionate about both technology and the outdoors. I enjoy climbing, skiing, hiking, trail running, and ski touring—activities that challenge me physically and mentally. As an amateur climber and alpinist, I find joy in pushing my limits and exploring new terrain. Time in the mountains helps me disconnect from technology and reconnect with nature, bringing balance and perspective to my life.

## PRIVACY

In compliance with the Italian Legislative Decree no. 196 dated 30/06/2003, I hereby authorize the recipient of this document to use and process my personal details for the purpose of recruiting and selecting staff and I confirm to be informed of my rights in accordance to art. 7 of the above-mentioned decree.

Trento, 21st May 2025

